



TRANSPARENCY OF ETHICAL PRODUCTS AND ITS SOURCE IN SUPPLY CHAIN

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Abstract - In this work the functioning of Blockchain technology and the possible use or impact it may have on current Supply Chain Management (SCM) Registry systems and the role of legal experts are described. The spread of Blockchains is an important aspect for anyone in the trust business.

The proposed system maintains the Blockchain base transactions which manages the SCM records and automatic data recovery from third party attacks, the system also addresses the issues of data inconsistency during the transaction. The main objectives were to define how Blockchain can change the supply chain and logistics industry and maintain a transparency of products between the consumer and manufactures in the whole process. The typical challenges in these spheres were considered and the main key features of Blockchain that can resolve these difficulties were marked. In survey we discovered possible challenges and benefits of Blockchain-based applications. Considering the current situation in the supply chain and logistics industry, this thesis can empower different businesses to start working with the companies that are creating block chain-based applications.

Keyword: Blockchain, Supply chain, source, transparency, products, Counterfeit product, QR code

I. INTRODUCTION

Blockchain is being termed as the fifth disruptive innovation in computing. In simplest words, it is a distributed ledger of records that is immutable and verifiable. Since its advent in 2008, Blockchain as a concept has been used in various ways. The largest impact or application is seen as a multitude of crypto currencies that have sprung up. However, with time, it is proven that Blockchain as a technology is likely to have an impact much wider than just the cryptocurrency domain and much deeper than simple Distributed ledger storage. This project implementation intends to bring together all the key developments research done in Blockchain. While the most common adoption of Blockchain is in finance and banking domain, there are experiments being conducted by numerous organisations in various domains.

Blockchain deployment outside finance has been largely experimental. Some of the most promising non-finance operations of Blockchain are anticipated to include those in supply chain. Among numerous conditioning that are likely to be converted by blockchain, supply chain therefore deserves special attention. An adding reliance on the application of Internet-of-Things (IoT) operations is among the trends which will affect Supply Chain Management operations (SCM). With IoT, Radio-Frequency Identification (RFID) markers, detectors, barcodes, GPS markers and chips, the locales of products, packages and shipping holders are frequently tracked at each step.

The problem of fake products, especially pharmaceutical products, has agonized the transnational community for decades. The World Health Organization (WHO) estimates that encyclopaedically, 10 percent of drugs are fake, rising to 30 percent in developing countries. The battle against counterfeiting remains a big challenge. This is of significant concern, as fake pharmaceutical products can have adverse effect on one and indeed death.

Over the last many times, Blockchain technology has been a seductive result for numerous different diligences beyond cryptocurrency. The logic behind this is frequently the translucency, security, quality assurance, global peer-to-peer deals, and decentralization that the blockchain provides. Unnaturally, a blockchain is a public, distributed tally that contains chained blocks, each of which is made up of several deals. The blocks are participated and accompanied across bumps via a peer-to-peer, distributed, and decentralized structure. Despite its implicit to elevate security, work and storehouse distribution, and translucency of force chains, there have been only many systems that examine the integration of anti-counterfeiting force chains and Blockchain technology. Due to the immaturity of Blockchain, it's inadequately understood and thus the intent to borrow it for force chain traceability is unknown. Likewise, despite the hype and possibilities, indeed the foremost innovative, technically superior technologies are rendered useless if they are not espoused by the druggies. For this reason, exploring and studying the problems of consumer technology relinquishment are bandied during a kind of disciplines. Within the marketing and logistics literature, Flint defines invention not as commodity new the earth, but as being new the stoner and latterly affirms a "voice of the client" approach to logistics invention and emphasis of the dynamic inter-organizational interface with guests.

II. LITERATURE SURVEY

2.1 Introduction:

Here we present a literature survey done to find out about numerous research done in the Blockchain in various domains. The use of Smart contracts, Distributed ledgers, usage of Blockchain nodes in privacy, security and transparency was learned and utilised in our project. Using Blockchain in various fields in one of the active research areas to explore in the world.

2.2 Research papers:

- 2.2.1 Smart Contracts [1]** also called crypto-contract, it is a computer program used for transferring and controlling the property or digital currents in specific parties. It does not only determine the terms and conditions but it also implements the policy / agreement. These smart contracts are stored on block-chain. Blockchain is an ideal technology to store these contracts due to the ambiguity and security. Whenever a transaction is considered, the smart-contract determines where the transaction should be transferred or should it be returned, owing to the fact that the transaction actually occurred.
- 2.2.2 Currently CSIRRO team has proposed a new approach to integrate Block on IOT with [2]:** In the initial stage, he uses smart-home technology to understand how IOT can be blocked. Block wheels are especially used to provide access control system for Smart Devices Transactions located on Smart-Home. Introducing Blockchain technology in IOT, this research again provides some additional security features; however, every mainstream Blockchain technology must have a concept that does not include and explain the concept of comprehensive algorithms. Besides, the technology used here cannot provide a general form of block-chain solution in case of IOT usage.
- 2.2.3 According to Ilya Sukhodolski [3]:** The AI system presents a prototype of multi-user system for access control over datasets stored in incredible cloud environments. Like other unreliable environments, cloud storage requires the ability to securely share information. Here the approach used provides access control over the data stored in the cloud without the provider's investment. Access Control Mechanism: the main tool which is the dynamic feature-based encryption scheme is used. Using Blockchain based decentralized badgers, the system provides an irrevocable log for accessibility requests for all significant security incidents like large financing, access policy assignment, alteration or cancellation. They also offer a set of cryptographic protocols that make the secret or secret key of cryptographic operation confidential. The hash code for the text of the sifter is only transmitted by a block on laser. The system has been tested on prototype smart contracts and tested on Ethereum Blockchain platforms.
- 2.2.4 Blockchain-Based IoT Application Using Smart Contracts: Case Study of M2M Autonomous Trading [4]:** This paper provides a basic IoT Blockchain fusion model with four layers which contains different types of IoT devices. Here they have used a Distributed file system which is considered in the model to store huge amount of IoT data. They have also provided a case study for blockchain-based IoT application, a Machine-to-Machine(M2M) autonomous trading system, which is further proposed on the Ethereum blockchain. They have built smart contracts for device registration, data storage, service provision and fair payment, and the proof-of-concept is implemented using two Raspberry Pi's to interact with smart contracts. The proposed system ensures that the blockchain can improve IoT applications transparency, tracking and security.
- 2.2.5 Edgence: A blockchain-enabled edge-computing platform for intelligent IoT-based Apps [5]:** Edgence (EDGE + intelligENCE), it is proposed system to serve as a blockchain-enabled edge-computing platform to intelligently manage massive decentralized Applications (dApps) in IoT use cases. Used to extend Blockchain's scope to IoT-based dApps, Edgence adopted master node technology to connect to blockchain-based systems in the real world. A master node contains a full node of the blockchain and a collateral, it is deployed on an edge cloud of mobile edge computing, which is convenient for the master node to use resources of the edge cloud to run IoT dApps.
- 2.2.6 HCloud: A trusted JointCloud serverless platform for IoT systems with Blockchain [6]:** This introduces HCloud, a trusted JointCloud platform for IoT systems using server less computing model. It allows an IoT server to be implemented with multiple serverless functions and schedules these functions on different clouds based on a schedule policy. The policy is then specified by the client and includes the required functionalities, execution resources, latency, price and so on. HCloud collects the status of each cloud and dispatches server less functions to the most suitable cloud based on the schedule policy. By leveraging the blockchain technology, it is noticed that the system can neither fake the cloud status nor wrongly dispatch the target functions.
- 2.2.7 Privacy preserving attribute- keyword based data publish-subscribe service on cloud platforms [7]:** They offer a secure electronic health record (EHR) system based on special-based crypt co-occurs and blockchain technology. In this system, there is a use of attribute-based encryption (ABE) and identity-based encryption (IBE) to encrypt medical data and to use identity-based signature (IBS) to apply digital signatures. In order to obtain various functions of ABI, IBE and IBS in crypto, they have presented a new cryptographic primitive, which is called a joint feature-based / identity-based encryption and signature (C-AB / IB-ES). It simplifies system maintenance and does not require the installation of separate cryptographic system for various security requirements. In addition, they have also used block connect techniques to ensure the integrity and inspection of medical data.

- 2.2.8 Random Seed Generation for IoT Key Generation and Key Management System Using Blockchain [8]:** Here a scheme is used to generate seed needed for key generation and a scheme to manage the public key using blockchain. First there is a random seed generation scheme needed for key generation. In order to prevent the risk of a man-in-the-middle attack and reverse engineering, seeds are generated by using out-of-band communication and hardware variation. Second is a key management system for the IoT using blockchain. The scheme proposed is to distribute the public key on the blockchain network. The public key is used to encrypt a session key that will be used for communication between devices

III. PROBLEM FORMULATION

3.1 Motivation:

In complex production, a large amount of data is generated which can be the source for value-add for actors in the value chain. Blockchain-based systems provide the potential to capture information along the production chain, to provide means for collaboration and they ensure integrity of information.

The worldwide improvement of an item or innovation consistently accompanies hazard factors, for example, forging and duplication also, use of harmful products. Due to awareness of cruelty free environment most users prefer to know the source of the products they are using.

Forging items can influence the organization's name and the client's wellbeing. Now-a-days discovery of forged item is the greatest test. Counterfeit and Environment harmful products are causing a significant impact on the organization and the client's wellbeing. Hence, the creators of products are defying enormous misfortune.

India and different nations are battling to create an environment friendly product.

3.2 Aim:

To design and develop a reliable, responsible, modernized, transparent, and secured supply chain against counterfeiting attacks. To ensure integrity of information and to benefit in revenue growth which will help in Economy development.

3.3 Problem Statement:

In common, users do not know sufficiently about the products that they buy and sell to navigate the much-complicated challenges faced in today's global supply chains (e.g., safe, sustainable, and ethical). Certain companies are getting to know about the business value of traceability for effectiveness, cost savings, and achieving product premiums in today's market. Still, they must first overcome the doubt associated with validating claims of product identity and traceability.

Blockchain technology was used to solve the problem of supply chain traceability, further addressing the safety issues, and to demonstrate its link in each supply chain in the implementation process details.

Further can be done to prepare companies with real-time traceability of products of global supply chains.

IV. PROPOSED SYSTEM ARCHITECTURE

This system highlights the implementation of e-transaction using Blockchain for such a proposal from a practical Point of view in both development of the project / deployment of the project and usage contexts. Concluding this work is a potential roadmap for Blockchain technology to be able to support complex applications. Building an electronic transaction system that satisfies the legal requirements of legislators has been a challenge for a long time. Distributed ledger technologies are an exciting technological advancement in the Information Technology world.

The proposed system uses Blockchain as a main domain. With the use of Blockchain Supply Chain Management, there will be transparency in supply chain as they will allow the supply chain nodes to verify the authenticity of a product's data. Data plays a crucial role in today's business world. In the traditional SCM the data regarding the product is manipulated and lost in the process so to overcome it a secure encryption and verification technologies are used which are present in Blockchain. This ensures that no data is manipulated and lost. Data stored in the blockchain database is immutable. It is very supportive if we want to keep the integrity of a data and make sure that nobody ever tampers with. There are significant advantages of adopting Blockchain in supply chain management: Customer Fulfilment, Anti-Counterfeit, Identify the problems, cost efficiency and rise in Economy. When we take into account all these factors, we are motivated to generate a system using Blockchain which maintains transparency between the consumer and the manufacturer/ distributor. The same is predicted in the **Fig. 4.1**, which depicts the design.

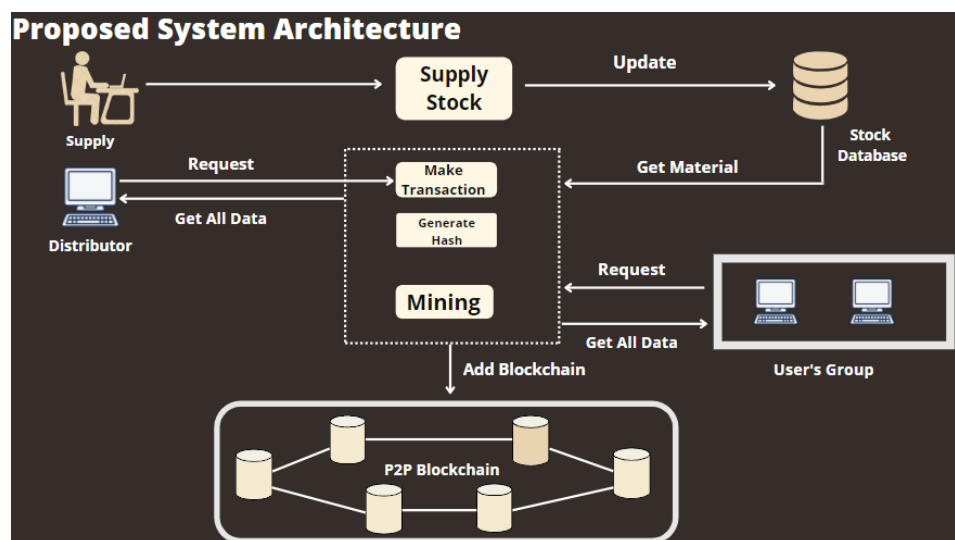


Figure 4.1: Proposed System Architecture

We proposed a system using blockchain technology as a web application for the tracing of products back to its source and for detection of counterfeit products. The proposed system ensures that the products bought are safe and sustainable right from its source which are used in day-to-day life. The proposed system as in the Fig 4.2 consists of three main parts, customer or user web application, Manufacturer's or Distributor's web application, and Database.

The first application is the Manufacturers or Distributor's application in which the distributor will have to first register themselves. After registration and login into the application, there will be list of option. The first one is to add a product in which the manufacturer can add the product details. Another option is to show the order in which they can see customers' order details and after that, they can decide the accept or reject the order. The manufacturer also can see whether the product is delivered or not.

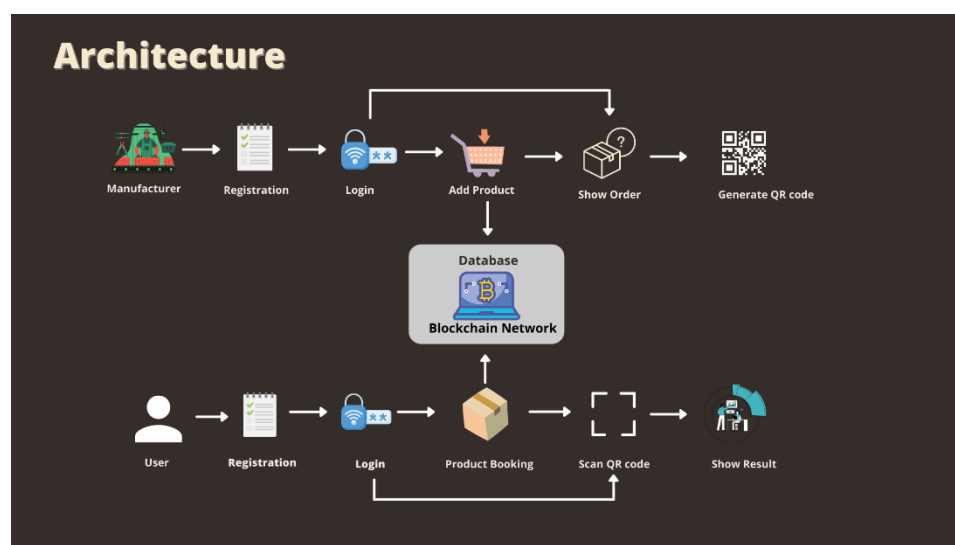


Fig 4.2 Architecture of the Application

A second application is the Customer application. For new users they have to first register in the application and creating their account. After successful registration, the user can log in to the application using username/ email address and password. In this app, there is an option to display products where customers can see product details like name, total price, product price, manufacturer details, production and expiration date of product. The user can book the product, after the user types in the quantity of the product required. In this application, we can view orders using my order where we can see product details, name, price, date, time, price, and status of the product whether it is delivered or not. After successful product placement, there is a unique QR code generated for the product. In this application we have a QR code scanner in which we scan the QR code of the product then it shows the source of the product. If the product received is Counterfeit product, the QR code will not be scanned and it will show that the product is a counterfeit product.

In this project for the user to place an order he has to login in the application. After login, the user must fill in the details for ordering the product and book the product. The order of the product can be shown to the manufacturer/ distributor. The Distributer has to decide whether the order placed is accepted or rejected. Once the manufacturer accepts the order, unique QR code of a product will be generated. Once an order of product is stored on the Blockchain network, a hash code is generated of that product and it is possible to maintain the transaction of the product. In the proposed system QR code is generated for a particular product which is unique. Customers scan the QR code on the product or package using the application where they have the option of QR code scanner. After scanning, we get the result of the product source. In the end, the Blockchain system

holds these product details along with a history of transactions to enable the tracking of the product along the distribution chain. All the product details, block name, a hash value is stored in the Blockchain database.

4.1 Module Description

The system contains following modules:

- ◆ Admin
- ◆ Make transaction
- ◆ Block Generation and blockchain validation
- ◆ Consensus Algorithm validation and block chain recovery
- ◆ Results Generation

The central outline of the proposed algorithm is the implementation of supply chain management distribution data storage using Blockchain. The system creates reliable communication between multiple companies without using any third-party interface.

- ◆ We are using the Hash production algorithm and Hash will be created for the given series.
- ◆ Before making any purchase, we use peer verification to verify the data.
- ◆ If any chain is invalid then it will recover or update on the current Blockchain server.
- ◆ This will verify until all nodes are verified and make a query.
- ◆ The mining algorithm is used to test the hash generated in question until a valid hash is produced.

V. ALGORITHMS:

5.1 SHA-256:

SHA 256 may be a part of the SHA 2 family of algorithms, where SHA stands for Secure Hash Algorithm. Published in 2001, it had been a joint effort between the NSA and NIST. The significance of the 256 within the name stands for the ultimate hash digest value, i.e., irrespective of the dimensions of plaintext/cleartext, the hash value will always be 256 bits.

The other algorithms within the SHA family are more or less almost like SHA 256. Now, check out knowing a touch more about their guidelines.

Algorithm 1: Hash Generation: SHA-256

Input: Genesis block, previous hash, data d

Output: According to the given data a hash H will be generated.

- **Step 1:** Input data as d
- **Step 2:** Apply SHA 256 from SHA family
- **Step 3:** CurrentHash= SHA256(d)
- **Step 4:** RetrunCurrentHash

5.2 Peer-to-peer (P2P):

Peer-to-peer (P2P) networks have evolved as a cost-effective and efficient way to distribute videos to a large number of people through the Internet. P2P streaming systems (P2PSS) are difficult to test, analyse, or verify due to their dynamic and heterogeneous character. Complex communication protocols are required for these systems, and these protocols must be verified.

Because the central server can soon become overburdened as the number of nodes grows, server-based solutions have a bandwidth scaling difficulty. Peer-to-peer systems are required because they offer a greater number of communication channels and hence scalability. Previously, verification of communication protocols was pursued, especially for security and cryptographic systems. The protocol must be tested to ensure that it does not cause issues like deadlocks or process hunger, and that it has correctness properties like the system gradually achieving specified operating states.

Algorithm 2: Protocol for Peer Verification

Input: User Transaction query, Current Node Chain CNode[chain], Other Remaining Nodes, Blockchain NodesChain[Nodeid] [chain]

Output: If any chain is invalid it will Recover, else execute the current query

- **Step 1:** User will generate the transaction using DDL, DML or DCL query
- **Step 2:** Get current server blockchain Cchain βCnode[Chain]
- **Step 3:** For each,

$$NodesChain [Nodeid, Chain] \sum_{i=1}^n (GetChain)$$

- End for
- **Step 4:** For each (read i into NodeChain)
If (!equals NodeChain[i] with (Cchain))
Flag 1
Else Continue Commit query
- **Step 5:** if (Flag == 1)
Count = SimilaryNodesBlockchian()
- **Step 6:** Calculate the majority of server
Recover invalid blockchain from specific node
- **Step 7:** End if
End for
End for

5.3 Mining Algorithm

A number of mining algorithms have been developed, each with its own set of strengths and disadvantages. The NSA-created SHA-256 is one of the most widely used algorithms.

A mining algorithm is a set of rules used by cryptocurrency miners to determine how to create new blocks.

Algorithm 3: Mining Algorithm for valid hash creation

Input: Hash Validation Policy $P[]$, Current Hash Values hash_Val

Output: Valid hash

- **Step 1:** System will generate the hash value (hash_Val) for i th transaction using Algorithm 1.
- **Step 2:** if (hash_Val .valid with $P[]$)
 - Valid hash
 - Flag =1
 - Else
 - Flag=0
 - Mine again randomly
- **Step 3:** Return valid hash when flag=1

VI. MATHEMATICAL MODEL

A System has represented by 5-different Phases, each phase works with own dependency System(S).

Here $S = (Q, \Sigma, \delta, q_0, F)$ where,

- Q is a finite set of states.
- Σ is a finite set of symbols which is called the alphabet.
- Δ is the transition function, where $\delta: Q \times \Sigma \rightarrow Q$
- q_0 is the first condition in which any input is processed ($q_0 \in Q$).
- F is a set of final state or states of Q ($F \subseteq Q$).

All (n) data nodes will return 1 when each have the same blockchain.

- Q = initial transaction data with genesis block
- $\Sigma = \{\text{SHA-256, Consensus_Val, Mining}\}$
- Δ = Validate all server ($S1 \subseteq S2 \subseteq S3 \subseteq S4$) all server validation process
- q_0 = Initial transaction $T[0]$
- $F = \{\text{Commit Trans, Get_History_Record}\}$
- State $\Rightarrow$ 1: if all chains are validated or same
0: if any $t(n)$ server consists the invalid chain

VII. RESULT

The time required for each algorithm to validate the Blockchain in four nodes is shown in Figure 6.1. The X axis depicts the algorithm used in the Blockchain, while the Y axis depicts the time needed in milliseconds for each of the four nodes for the given algorithm



Fig 6.1 Time required for the algorithms in milliseconds

The time required for the consensus algorithm to validate the Blockchain in four nodes is shown in Figure 6.2. The X-axis depicts the algorithm process of the Blockchain for each node, while the Y-axis depicts the time needed in milliseconds for each of the four nodes.

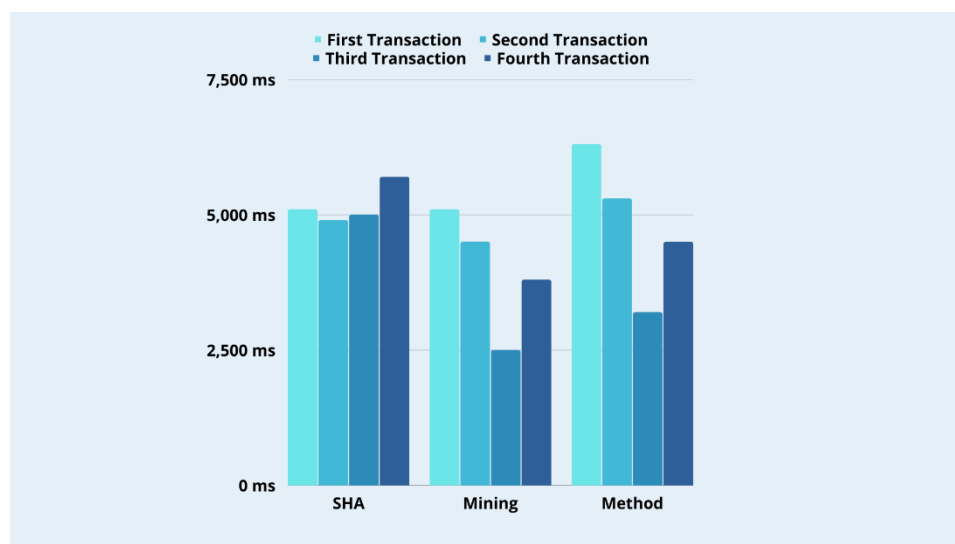


Fig 6.2 Time required to validate the Blockchain in the four nodes

The QR code generate is highly encrypted and is unique for each and every product. The QR codes data will be shown only through the application scanner. Due to high level of security and encryption in the Blockchain nodes. The data stored cannot be tampered with, maintaining transparency between the consumer and manufacturer which in turn helps in building the trust between them. The below QR Code in Fig 6.3 is generated by the system after a successful order placement. The QR Code is only scannable only through the system, it won't generate any data after scanning through other system or application. After successful scanning, the system will show the user the details and source of the product.



Fig. 6.3 QR Code generated by the system

CONCLUSION

Blockchain is a distributed platform that works without centralized authority and allows for the creation of a separate and uncluttered directory for verified and trackable activities. Because of these unique features, it has been tested for different applications, especially in the financial sector. This paper makes contributions not only to the science textbook blockchain in the supply chain management domain but also to the T&C industry sector. Blockchain is continuous. First, the full requirements of the operating system have been identified. Then, we introduced a comprehensive protocol that allows each team, including supply chain partners and customers. Counterfeit products are growing at an alarming rate online. Therefore, there is a strong need to obtain counterfeit products and blockchain technology is used to obtain counterfeit products. In addition, the information is encrypted with QR code. Customers or users scan the QR code and be able to identify the fake product. Digital product details can be stored in the form of blocks in blockchain technology. Therefore, the proposed system is useful for the customer to find counterfeit products in the supply chain. Customers can scan the generated QR code and can get all the information as to whether the product is genuine or not.

FUTURE SCOPE

As evidenced by the growing number of successful pilot projects, blockchain technology can improve supply chain management by providing visibility and reliability of action information between participating parties. The assurance, transparency, evidence of disruption, and trust provided by the blockchain can help make transaction data easier to share and should provide a platform to market responsible processes for all participants in the supply chain with greater transparency, thus making it easier for hold teams to respond. For example, BeefLedger has integrated blockchain and Internet of Things (IoT), statistics, smart contracts, and digital tokens to track availability, pay payments, reduce fraud risk, and provide transparency to cattle breeding in Australia. The future scope of this work uses the proposed method on the sub-control platform to develop a concept-proof prototype to automatically detect Bar / QR codes.

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